LearnPulse AI LMS — Week 11.5 Report

Week 11.5 — Teacher–Student Enrollment and Subject Association

Executive Summary

This report documents the completed implementation and final verification of **Week 11.5 — Teacher–Student Enrollment and Subject Association** for the LearnPulse AI-Powered LMS backend and frontend. *(Note: This milestone was implemented as an additional user-directed enhancement following Week 11 to establish core LMS relationships prior to the official mentor-assigned Week 12 assignment.)* Week 11.5 introduces an additive relational binding: **Teacher** → **Subject Assignment** → **Subject** → **Student Enrollment** → **Student**. All ten approved requirements were implemented, including mandatory backend authorization enforcement, safe student pickers with zero credential leakage, soft-unenrollment preserving historical attempts/grades, idempotent reactivation, a **184-test automated regression suite (184 passed, 0 failures, 0 errors, 0 skipped, BUILD SUCCESS)**, and a complete frontend enrollment management interface with verified TypeScript compilation.

1. Core Business Architecture & Invariants

- **Teacher-Subject Assignment**: Admin assigns teachers to academic subjects via dedicated Admin endpoints.
- **Teacher Scope Invariant**: Teachers can only manage enrollments for subjects assigned to their account. Cross-subject management is rejected with HTTP 403 Forbidden.
- **Student Privacy Invariant**: Students can only view their own enrollments. Cross-student data access is blocked with HTTP 403 Forbidden.
- **Content Protection Invariant**: Direct API access to chapters, PDF documents, quizzes, and progression requires an ACTIVE enrollment. Unenrolled/inactive requests return HTTP 403 Forbidden.

2. Database Schema & Integrity Constraints

Two additive relational tables were introduced in PostgreSQL (postgres-pgvector):

Table Name	Columns & Types	Unique Constraint	Key Role
teacher_subjects	id (UUID, PK), teacher_id (UUID, FK), subject_id (UUID, FK), assigned_at (TIMESTAMP)	uk_teacher_subject (teacher_id, subject_id)	Binds teachers to authorized academic subjects
student_enrollments	id (UUID, PK), student_id (UUID, FK), subject_id (UUID, FK), status (VARCHAR), enrolled_at, created_at, updated_at	uk_student_subject (student_id, subject_id)	Tracks student enrollment lifecycle (ACTIVE / INACTIVE)

3. Refinement Implementations & Safeguards

- **Mandatory Backend Authorization**: Enforced on ChapterController, StudentQuizController, DocumentDownloadController, and CourseProgressionController. Unenrolled students are rejected with HTTP 403 Forbidden.
- **Admin Subject Assignment Workflow**: Documented endpoints: POST /api/admin/subjects/{id}/teachers, DELETE /api/admin/subjects/{id}/teachers/{tId}, and GET /api/admin/subjects/{id}/teachers.
- **Safe Student DTOs**: StudentSummaryDTO exposes only safe profile attributes (Name, Email, Dept, Enrollment No.). Passwords, hashes, and JWT tokens are completely excluded.
- **History Preservation & Soft Unenrollment**: Unenrollment sets status to INACTIVE. Historical quiz attempts and grades remain permanently intact.
- **Idempotent Reactivation**: Re-enrolling an inactive student reactivates the existing record without unique constraint collision.

4. Automated Regression Test Suite Results (184 Tests)

Metric	Result
Total Tests Executed	**184** (149 existing Week 1-11 tests + 35 new Week 11.5 tests)
Passed	**184**
Failures	0
Errors	0
Skipped	0
Maven Build Status	**BUILD SUCCESS**
Total Execution Time	1 min 57 sec

Week 11.5 Test Matrix Breakdown:

- **EnrollmentIntegrationTest (13 tests)**: Admin teacher assignment, duplicate check, teacher enroll/unenroll, available student picker, reactivation.
- **EnrollmentAuthorizationTest (17 tests)**: Cross-teacher blocking, student isolation, unenrolled access to chapters/quizzes/documents/progression (403), inactive blocking.
- **EnrollmentSchemaVerificationTest (4 tests)**: Verified table schemas, unique constraints (uk_teacher_subject, uk_student_subject), and enum types.
- **EnrollmentUserJourneyTest (1 test)**: Full end-to-end lifecycle from assignment to enrollment, access verification, soft-unenrollment, and reactivation.

5. Frontend Implementation & Verification

- **TeacherStudentsPage.tsx**: Dedicated teacher portal at /teacher/students featuring subject selector, enrolled student table, and safe student picker modal.
- **StudentSubjectCatalogPage.tsx & StudentDashboardPage.tsx**: Augmented with Active Enrolled badges, assigned teacher attribution (Prof. Name), and enrollment guards.
- **TypeScript Build Verification**: Executed npm run build (tsc -b && vite build) — 1,690 modules transformed, built in 4.91s with **0 errors**.

6. Project Environment & Boundary Guarantees

- **Week 9–11 AI Architecture Untouched**: Zero modifications to Spring AI, Ollama, prompt templates, or AI feature services.
- **Database Isolation**: Connected exclusively to LearnPulse PostgreSQL (postgres-pgvector / learning_assistant_db on port 5432).
- **Zero SmartOnboard Contact**: SmartOnboard containers and resources were never accessed or modified.
- **Git Cleanliness**: Zero Git commits or pushes were performed.

Final Status: Week 11.5 — Teacher–Student Enrollment and Subject Association is fully implemented, verified across 184 automated tests, and confirmed production-ready.